

Backtracking

try all the choices!

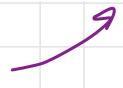
splitting text $A[1..n]$ into words

PRIMVS | DIGNITAS | IN | TAM | TEN | VIS | SCIENTIA | NON | POTEST |
ESSERE | SENIM | SVNT | PARVA | E | PROPE | IN | SINGVLIS | LITTERIS
ATQVE | INTER | PVNCTIO | NIBUS | VERBORVM | OCCVPATAE

choice: what is the next word?

ask: how would you do this?
identify 1st word,
then try to split the rest

BLUE	STEM	UNIT	ROBOT	HEARTHANDSATURNSPIN
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how can I split  into words?

it's however I can split  into words

BLUE	STEM	UNIT	ROBOT	HE	ARTHANDSATURNSPIN
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or however I can split  into words

BLUE	STEM	UNIT	ROBOT	HEAR	THANDSATURNSPIN
------	------	------	-------	------	-----------------

or however I can split into words

BLUE	STEM	UNIT	ROBOT	HEART	HANDSATURNSPIN
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or however I can split into words

BLUE	STEM	UNIT	ROBOT	HEARTH	ANDSATURNSPIN
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union of these is going to be splittable into words?

$\text{Splittable}(A[1..n]) :$

if $n = 0 :$

return T

for j in 1 to $n :$

if $\text{isWord}(A[i..j]) :$
if $\text{splittable}(j..n) :$
return T

return F

write this 1st

ask: how do I write the choices?

but we don't want to pass the whole array around...

Define $\text{splittable}(i) = \text{T}$ iff suffix $A[i..n]$ is splittable

how do I implement splittable?

$\text{splittable}(i) :$

if $i > n :$

return T

else:

for j in i to n:

if isword(i, j):

if spittable(j+1):

return T

return F